SUT GLIWICE
INSTITUTE OF COMPUTER SCIENCE

IoT Internet of Things

Laboratory manual

Table of Contents

Introduction 3
IoT Raspberry Pi device 3
Raspberry Pi OS systems 6
Windows 10 IoT Core Instalation 9
To Do 21
Bibliography 21

Introduction

The aim of this exercise is to present the latest industrial solutions to provide Internet of Things developing.

The internet of Things (IoT) is the network of physical objects—devices, vehicles, buildings and other items—embedded with electronics, software, sensors, and network connectivity that enables these objects to collect and exchange data. The IoT allows objects to be sensed and controlled remotely across existing network infrastructure, creating opportunities for more direct integration of the physical world into computer-based systems, and resulting in improved efficiency, accuracy and economic benefit when IoT is augmented with sensors and actuators, the technology becomes an instance of the more general class of cyber-physical systems, which also encompasses technologies such as smart grids, smart homes, intelligent transportation and smart cities. Each thing is uniquely identifiable through its embedded computing system but is able to interoperate within the existing Internet infrastructure. Experts estimate that the IoT will consist of almost 50 billion objects by 2020.

British entrepreneur Kevin Ashton first coined the term in 1999 while working at Auto-ID Labs (originally called Auto-ID centers, referring to a global network of objects connected to radio-frequency identification, or RFID). Typically, IoT is expected to offer advanced connectivity of devices, systems, and services that goes beyond machine-to-machine (M2M) communications and covers a variety of protocols, domains, and applications. The interconnection of these embedded devices (including smart objects), is expected to usher in automation in nearly all fields, while also enabling advanced applications like a smart grid, and expanding to the areas such as smart cities. "Things," in the IoT sense, can refer to a wide variety of devices such as heart monitoring implants, biochip transponders on farm animals, electric clams in coastal waters, automobiles with built-in sensors, DNA analysis devices for environmental/food/pathogen monitoring or field operation devices that assist firefighters in Search and Rescue operations. Legal scholars suggest to look at "Things" as an "inextricable mixture of hardware, software, data and service". These devices collect useful data with the help of various existing technologies and then autonomously flow the data between other devices.

IoT Raspberry Pi device

DESCRIPTION

The Raspberry Pi is a low cost, credit-card sized computer that plugs into a computer monitor or TV, and uses a standard keyboard and mouse. It is a capable little device that enables people of all ages to explore computing, and to learn how to program in languages like Scratch and Python. It’s capable of doing everything you’d expect a desktop computer to do, from browsing the internet and playing high-definition video, to making spreadsheets, word-processing, and playing games.

What’s more, the Raspberry Pi has the ability to interact with the outside world, and has been used in a wide array of digital maker projects, from music machines and parent detectors to weather stations and tweeting birdhouses with infra-red cameras.

Several generations of Raspberry Pi's have been released. The first generation (Pi 1) was released in February 2012 in basic model A and a higher specification model B. A+ and B+ models were released a year later. Raspberry Pi 2 model B was released in February 2015 and Raspberry Pi 3 model B in February 2016. These boards are priced between 20 and 35 US$. A cut down compute model was released in April 2014 and a Pi Zero with smaller footprint and limited IO (GPIO) capabilities released in November 2015 for 5 US$.

All models feature a Broadcom system on a chip (SOC), which includes an ARM compatible CPU and an on chip graphics processing unit GPU (a Video Core IV). CPU speed ranges from 700 MHz to 1.2 GHz for the Pi 3 and on board memory range from 256 MB to 1 GB RAM. Secure Digital SD cards are used to store the operating system and program memory in either the SDHC or Micro SDHC sizes. Most boards have between one and four USB slots, HDMI and composite video output, and a 3.5 mm phono jack for audio. Lower level output is provided by a number of GPIO pins which support common protocols like I2C. Some models have an RJ45 Ethernet port and the Pi 3 has on board WiFi 802.11n and Bluetooth.

Fig. 1 ***Raspberry Pi board***

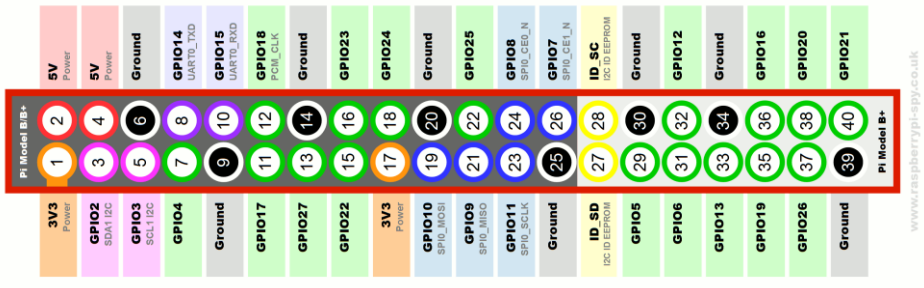

Fig. 2 Raspberry Pi II GPIO Layout diagram

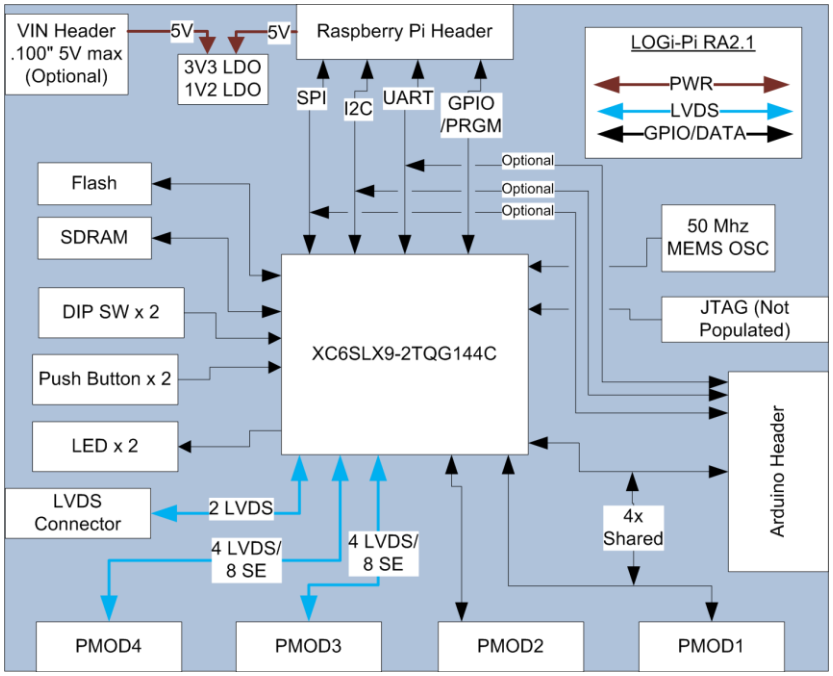

Fig. 3 Raspberry Pi block schematic

Raspberry Pi OS systems

DESCRIPTION

The most popular OS systems available for the Raspberry Pi boards are present on Fig.4.

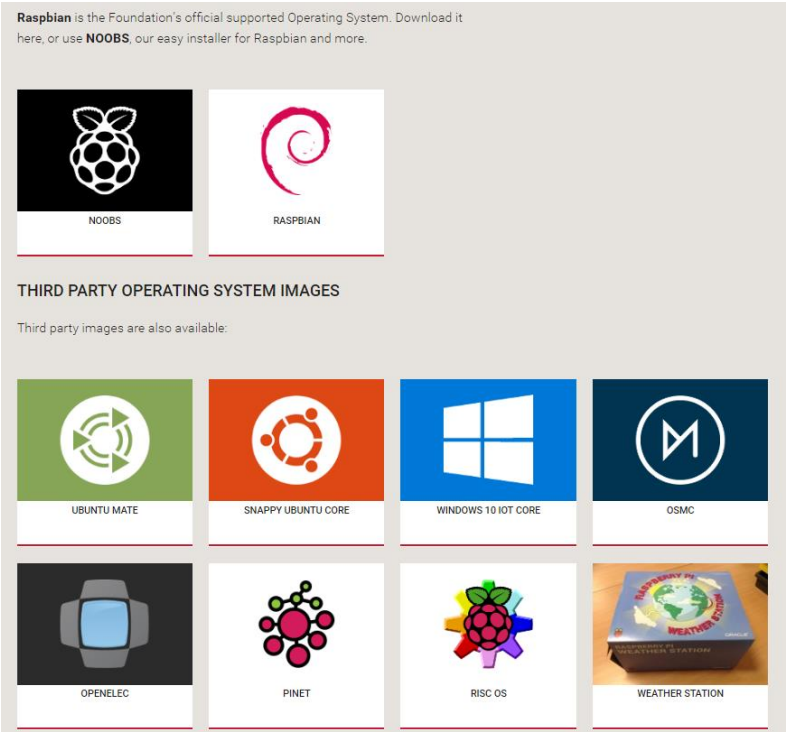

Fig. 4 Raspberry Pi available OS systems

Raspbian has been the standard Raspberry Pi operating system, more commonly referred to as a "distro", since the Pi arrived in 2012 and we have seen it grow over time into the capable distro that we use today. Raspbian is a fork of the massively popular Debian distribution and it is jointly maintained by the Raspberry Pi Foundation and the community. Being based on Debian, Raspbian comes with the APT (Advanced Packaging Tool) as it's package manager, which is used to install software from the vast Raspbian repositories, but Raspbian also comes with raspi-config, a menu based tool that simplifies the act of managing Raspberry Pi configurations such as setting up an SSH, overclocking and enabling the official Raspberry Pi camera. Since December 2014 the Raspbian desktop has received a notable improvement to its user interface thanks to the hard work of Simon Long. Moving away from the cluttered LXDE menu system we now have a simplified and refined interface that groups applications and configuration tools into clearer categories. As Raspbian is the default distro for Raspberry Pi it is also the distro that sees the most improvement and innovations examples of this are the RPi.GPIO library that enables Python to talk to the GPIO (General Purpose Input Output) pins, Minecraft the popular survival / sandbox game that has seen a massive investment of time from the community to enable a Python API for programming the game world. Lastly we have Sonic Pi the Ruby powered music creation application that has risen in popularity thanks to the efforts of Sam Aaron. With the release of the Raspberry Pi 2 we have seen Raspbian receive the speed boost that it always needed, by no means was it a slouch on the original Raspberry Pi, but the addition of more RAM and better CPU has enabled it to become a usable desktop distro that can easily be used as a main machine. Raspbian has improved since it was first released in 2012 and with each new release we see more refinements added to it, no wonder the majority of projects around the world are based upon this distro. It is a rock solid development base that runs well on the original Raspberry Pi but screams ahead on the new Raspberry Pi 2 specification.

Windows 10 IoT Core is a version of Windows 10 that is optimized for smaller devices with or without a display, and that runs on the Raspberry Pi 2 and 3, Arrow DragonBoard 410c & MinnowBoard MAX. Windows 10 IoT Core utilizes the rich, extensible Universal Windows Platform (UWP) API for building great solutions. Windows 10 IoT Core (Fig. 5) brings the power of Windows to your device and makes it easy to integrate richer experiences with your devices such as natural user interfaces, searching, online storage and cloud based services.

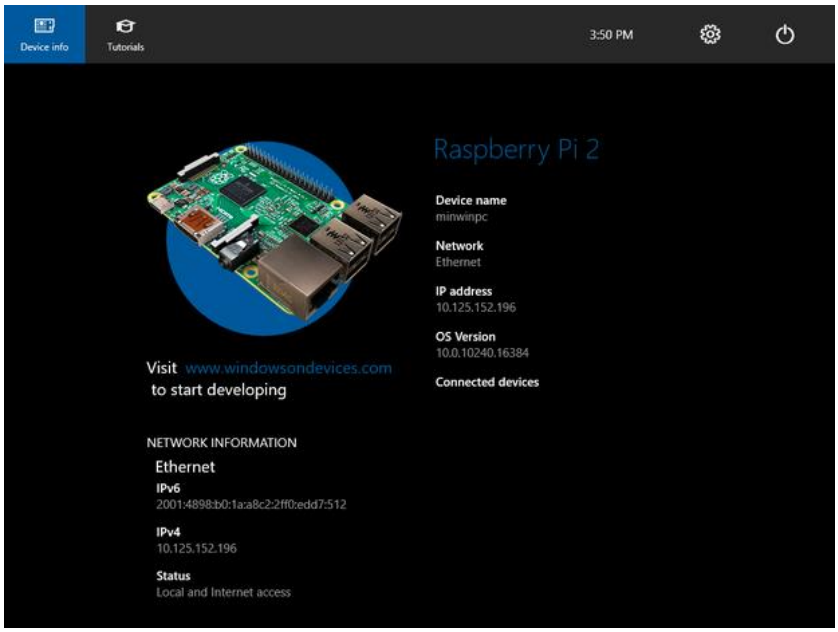

Fig. 5 Windows 10 IoT Core

WINDOWS 10 DEVELOPER TOOLS

All tools are available on Microsoft Developer web site:

https://developer.microsoft.com/en-us/windows/iot

Tools for Windows 10 IoT developing:

- Windows 10 IoT Core Dashboard http://go.microsoft.com/fwlink/?LinkID=708576
- Visual Studio 2015 environment https://go.microsoft.com/fwlink/p/?LinkId=534599
- Windows 10 IoT Core Release Image http://go.microsoft.com/fwlink/?LinkId=691711
- Windows 10 IoT Core Insider Preview http://go.microsoft.com/fwlink/?LinkId=733603
- Windows 10 IoT Image Helper
- Windows 10 IoT Remote Client https://www.microsoft.com/en-us/store/apps/windows-iot-remote-client/9nblggh5mnxz

OVERVIEW of the Raspberry Pi 3

Windows 10 now offers the full suport of the newest Raspberry Pi 3 board which has following benefits:

- Functional parity with RPi2, including Windows as a service updating and flighting model and a single FFU for RPi2 and RPi3
- Open sourcing Pi BSP to encourage community innovation/speed and support commercialization efforts
- Onboard WiFi drivers are now enabled for the Raspberry Pi 3
- UART/Serial (miniUART) driver for the Raspberry PI 3 onboard serial is now enabled for the Raspberry Pi 3
- Pin muxing support, increasing the number of available GPIO pins from 17 to 24.

- An additional SPI bus (SPI1).
- Increased UART baud rate support (was 115200, now 921600).
- 20% increase in GPIO performance

Windows 10 IoT Core Instalation

Installing the Windows 10 IOT Core

Microsoft Windows 10 IOT OS system is available for download from Windows Insider Preview Downloads.

Step 1

Register into the Microsoft Insider Program. To download images of the Microsoft IOT Core user must be logged into Microsoft Insider Program web page.

Step 2

On the download page User must choose which OS edition he wants to use in his project ieg. Windows 10 Iot Core Insider preview - build 17744 or 17733. The core numbers are changing depending on the latest developer editions available in the Microsoft repository. Microsoft IOT development policy is straightly tied with the latest Visual Studio environment. To fully use its power users are asked to use latest Visual Studio and Windows 10 IOT core builds in the development process synchronously. The Windows 10 IOT Core platform is still under development and improvement.

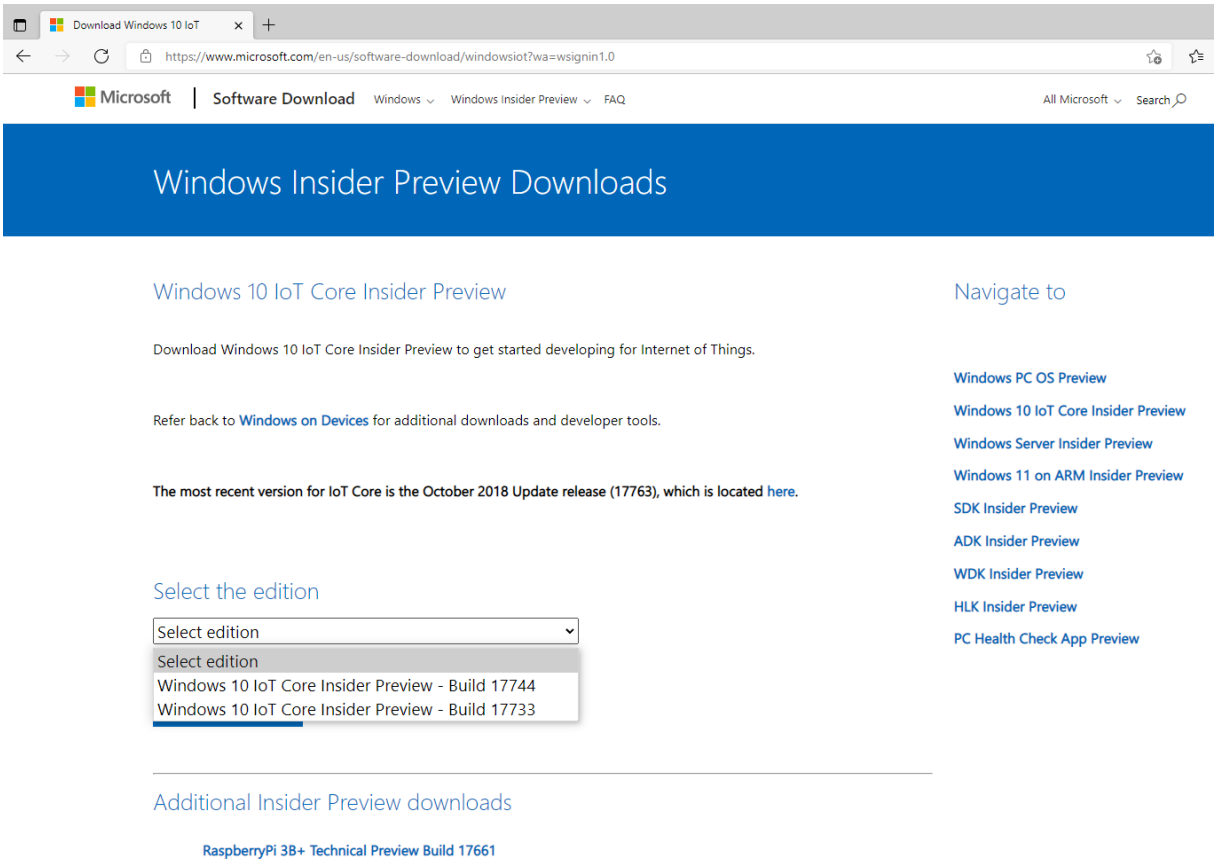

Figure 12: Windows Insider Program Win10 Core download page

Step 3

Insert the SD card to your computer SD cards reader slot,

Step 4

Run the Windows IOT Dashboard and follow its instructions - select the SD card drive letter, select the right FFU image, set the board name and administrator password

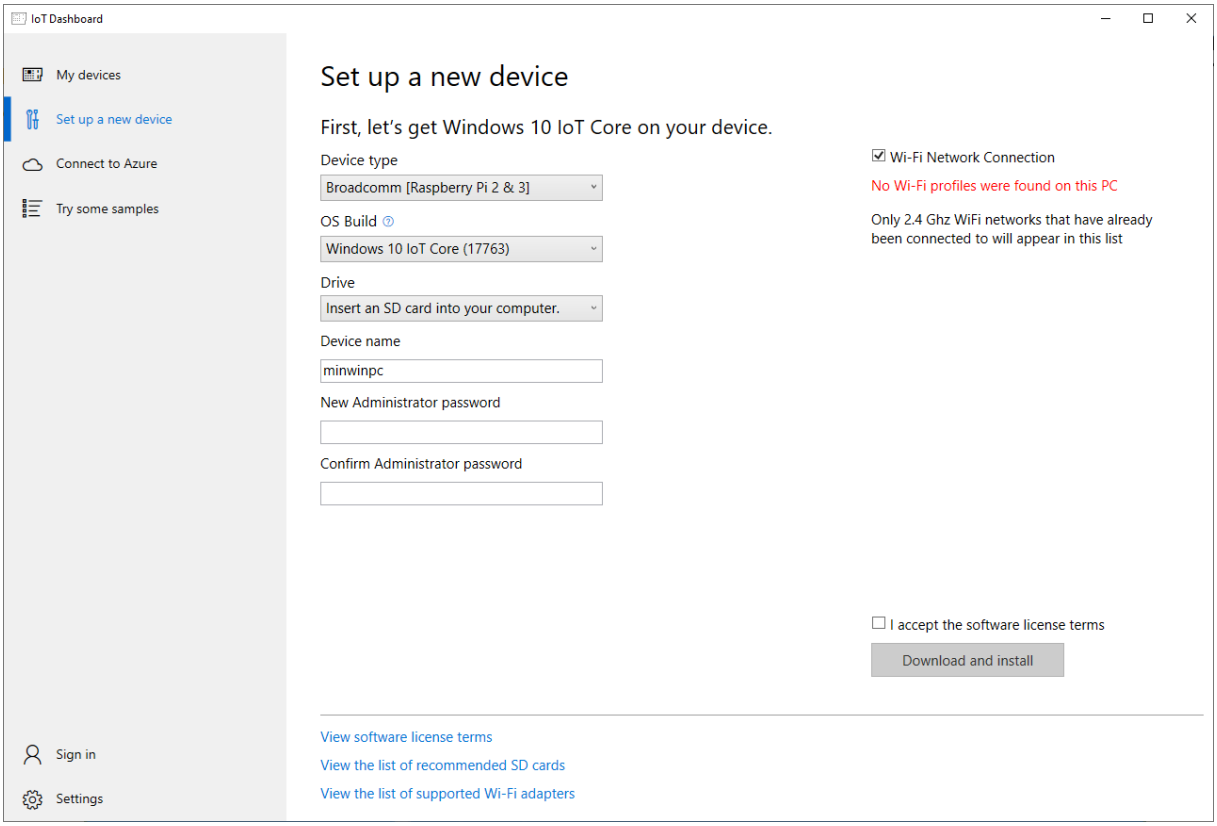

Figure 13: Windows IoT Dashboard view

Step 5

Gently remove SD card from the reader and push it into the Raspberry Pi SD card slot.

Step 6

Power on the Raspberry Pi board and follow the Windows 10 Core setup commands configuring the Windows 10 Core features. After the board reboot the Main Windows 10 Core screen displays:

Figure 14: Windows 10 Core system view

Raspberry Pi Programming Guide

This chapter describes the typical programming technics used in Raspberry Pi boards developing projects.

Raspberry Pi under Windows 10 IOT Core

To create and develop control applications on the Raspberry Pi boards needs the following development environment:

1. PC with Windows 10 System installed,
2. Visual Studio 2015 or higher,
3. Raspberry PI 2 or 3 board with Windows 10 IOT Core installed,
4. Configured TCP/IP network for Raspberry Pi and Windows 10 Desktop computer (Local LAN or Wi-Fi subnet),

Programming skills needed:

1. C# language knowledge,
2. XML/XAML language knowledge,
3. Windows API understanding.

For better development Raspberry applications, the Windows IoT Remote Client is welcome. This application is available for download from Microsoft Store. This application captures keyboard, mouse and screen from the Raspberry Pi board running the Windows 10 IOT Core system on the desktop PC. It allows developers to use standalone Raspberry board without a connected mouse, keyboard and monitor to it.

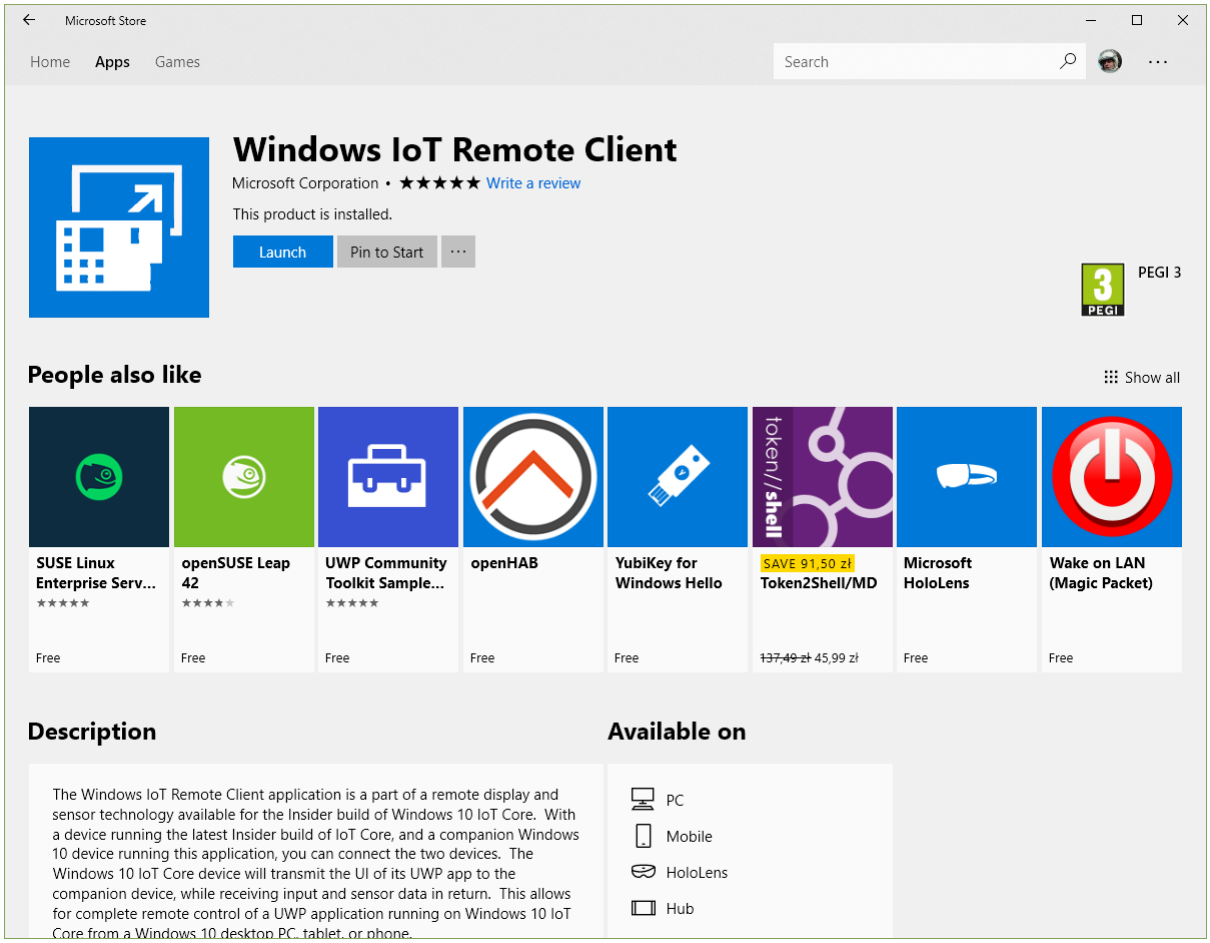

Figure 15: Microsoft Store - Windows IoT Remote Client

To write and develop applications under Windows 10 IOT Core developer must possess basic knowledge of how Windows operating system interacts with User applications. The major advantage of using Windows 10 IOT Core is that Microsoft concept based on use the same Kernel API available on different hardware platforms - desktop PCs, IOT boards suitable to run Windows Core, Tablets, phones etc. It reduces development costs due to the unifying system environment, and the only difference is in display view of same application code written in C#/C++. Windows 10 Core is specially designed to handle applications working as standalone on the IoT platforms in 24/7 time model.

Configuring the Windows 10 IOT Core platform

After installing the Windows 10 IOT Core, the developer must configure IoT platform using Windows Device Portal (WDP).

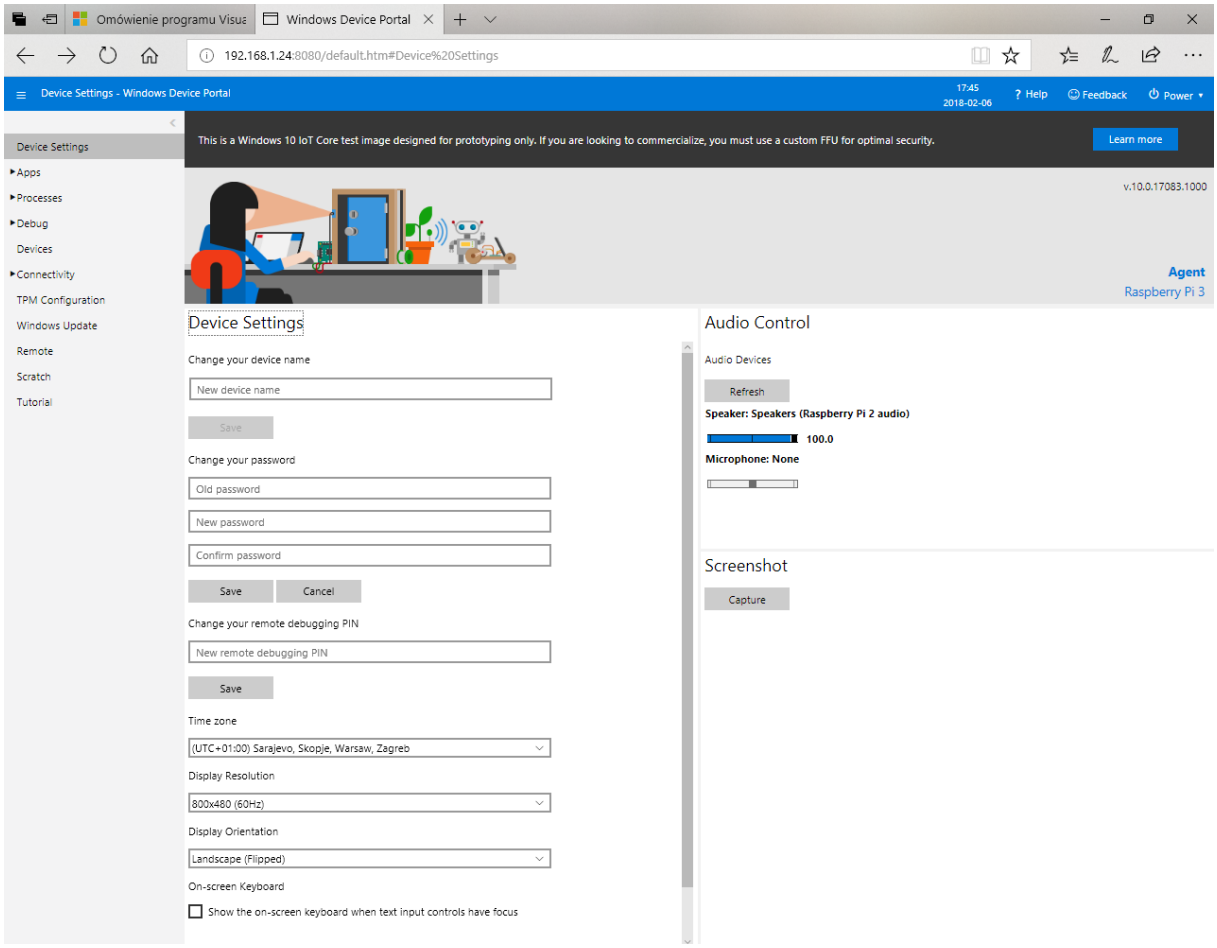

Figure 16: Windows Device Portal view

IoT board can be managed using any Internet browser - Chrome, Microsoft Edge, Firefox etc. To open the WDB portal on the IOT board user must enter the board IP address - *IPaddress:8080/default.htm*. The site is protected with username/password. Default account credentials are: **administrator/p@ssw0rd**. Following tabs in the WDP, it is possible to configure all necessary IoT platform settings, check the current board status, download development crash/debug information, configure network/Bluetooth settings, download drivers and configure security TPM modules. If all tasks are ready, the developer can start to write his own IoT application under Windows 10 IOT Core. Following steps are recommended before IoT board will be used for application:

Step 1

In the Device Settings user is recommended to Change your device name. The default name is minwinpc. The aim to change it is that if a user uses many of the IoT devices connected to the same network segment, it is difficult to recognise which role each device is set for. Enumerating IoT devices will show boards with the same name but with different IP addresses. Proper naming will make it easy to know what role each device plays.

Step 2

Because RPI boards don't have their own RTC clock modules Windows 10 IOT Core sets the time using the NTP services during its work. So very important in the industrial implementations and in a case when the time is important in developed applications is to set the proper time zone for the board. In the Device settings user is recommended to select proper Time zone

Step 3

Security reasons - the default password for the newly flashed device is p@ssw0rd. It is strongly recommended right after the first board boot to change it - to set it unique! It will prevent the IoT device from remote hacking. The password can be changed in Device Settings tab.

Step 4

The Windows 10 IOT Core comes with Cortana service ready. If the board is equipped with microphone and speakers, it is always possible to turn the Cortana service on for voice commands communication with the board.

Step 5

If the IoT board needs special hardware connected to it, then in the **Devices/Device Manager** user can upload and install appropriate driver for it in case if it is not preinstalled in the IoT Core.

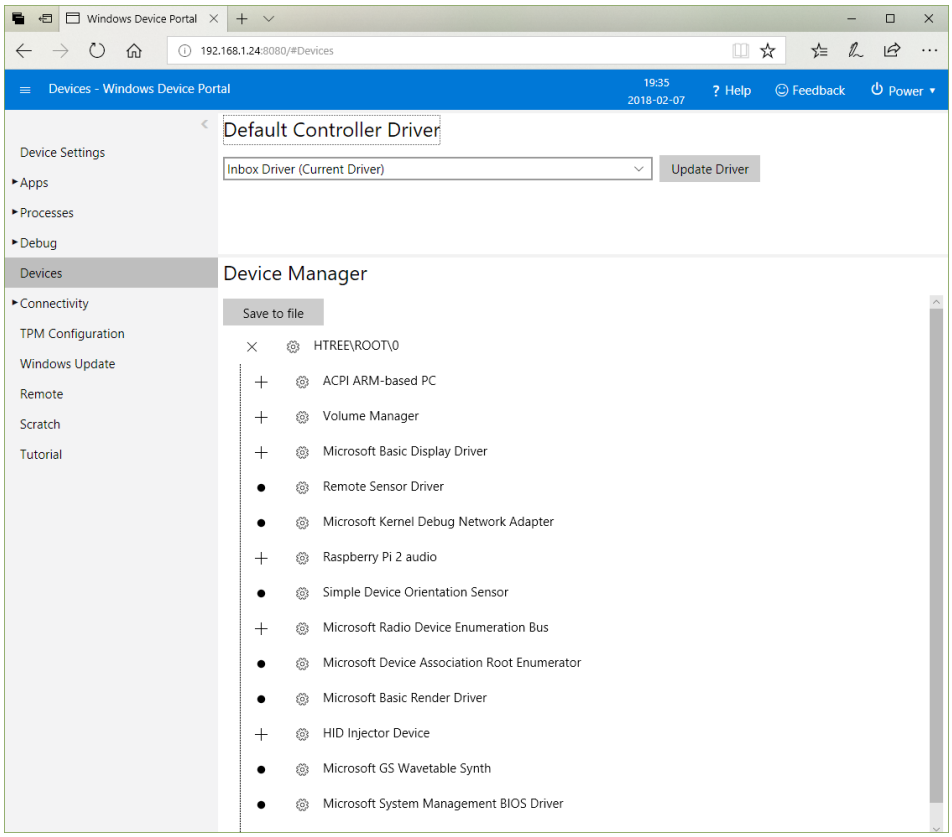

Figure 17: Device Manager view

Step 6

Raspberry Pi boards 1/2/3 are equipped with network connection modules. In case if the board under Windows 10 IOT Core is connected to LAN RJ45 connector the IP number can be set via DHCP server. In case if user wants to use Wi-Fi connection or to activate 4. IoT Hardware overview 224 Bluetooth then he can do it directly on the board main display or manage them via Windows Device Portal.

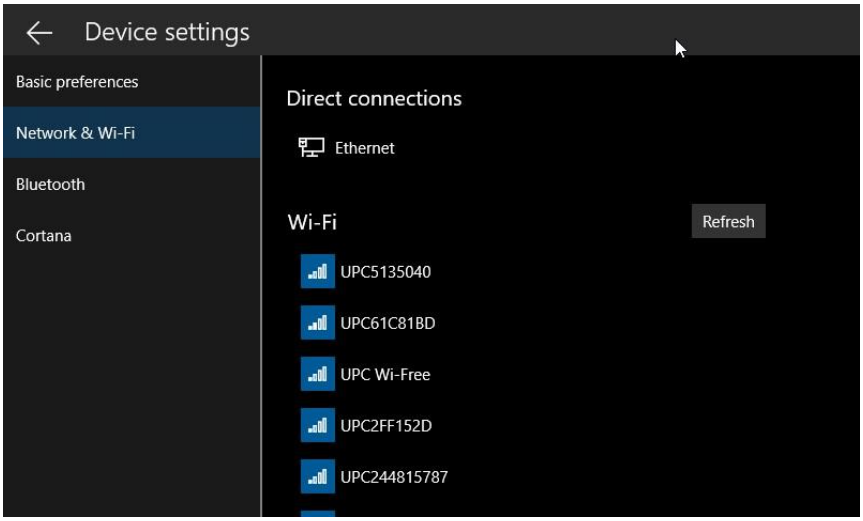

Figure 18: Network& Wi-Fi on the LCD view

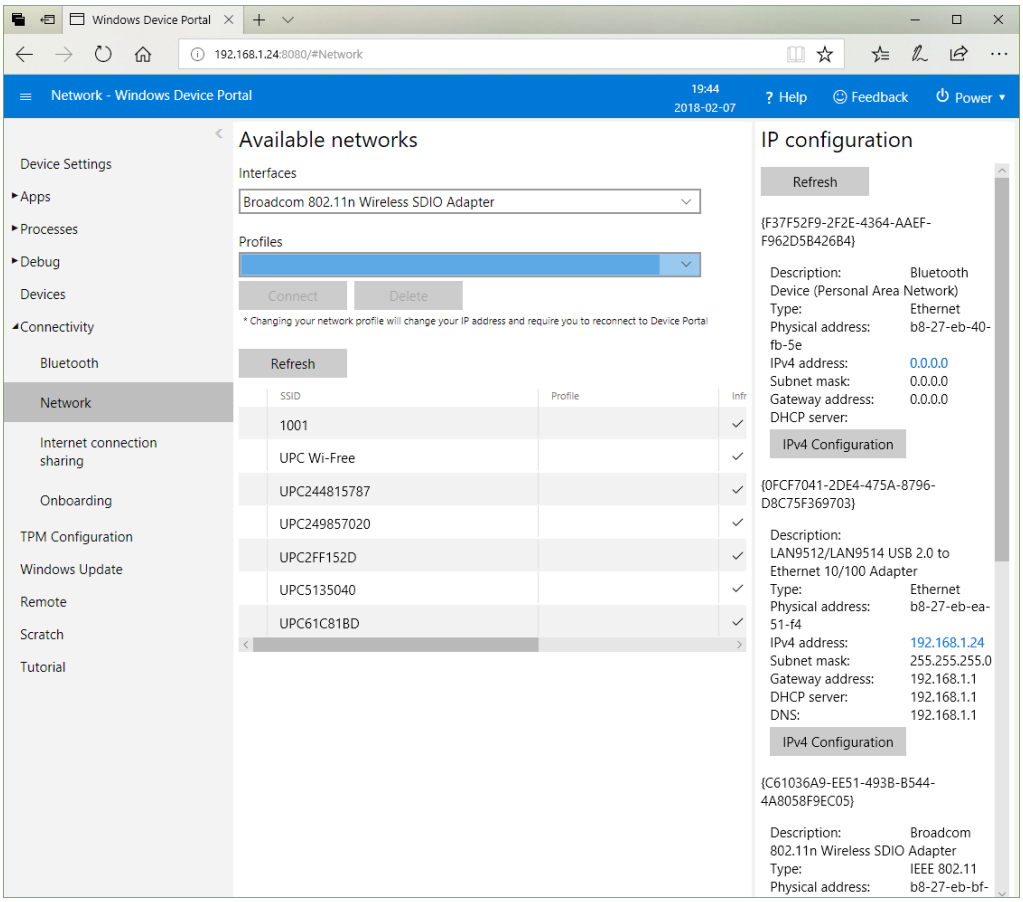

Figure 19: WDP Network view

Step 7

Security. In a case when IoT device must be protected for remote hacking one of solutions is to use Trusted Platform Modules (TPM) module following ISO/IEC 11889 standards for a secure cryptoprocessor, a dedicated microcontroller designed to secure hardware through integrated cryptographic keys. The chip contain physical security mechanisms to protect it from tampering, and malicious software is unable hack the TPM security functions. Some of the TPM key advantages are:

- Generate and store the cryptographic keys.
- Use the TPM unique RSA key technology for platform device authentication, which is burned into the chip.
- Help ensure platform integrity.

The most common TPM functions are used for system integrity measurements, key creation and use. During the system boot process, the boot code is loaded (including firmware and OS components) and can be measured and recorded in the TPM module. The integrity measurements are used for evidence how OS started and to be sure when correct boot software was used with TPM-based key. Windows 10 IoT Core supports few TPM modules standards which can be connected to the 40-pin GPIO connector.

Figure 20: TPM module view

Under the TPM Configuration tab in the Windows Device, Portal user can select the right communication protocol for the TPM module installed in the Raspberry Pi board. Then appropriate driver for the TPM module can be installed in the Device Manager tab.

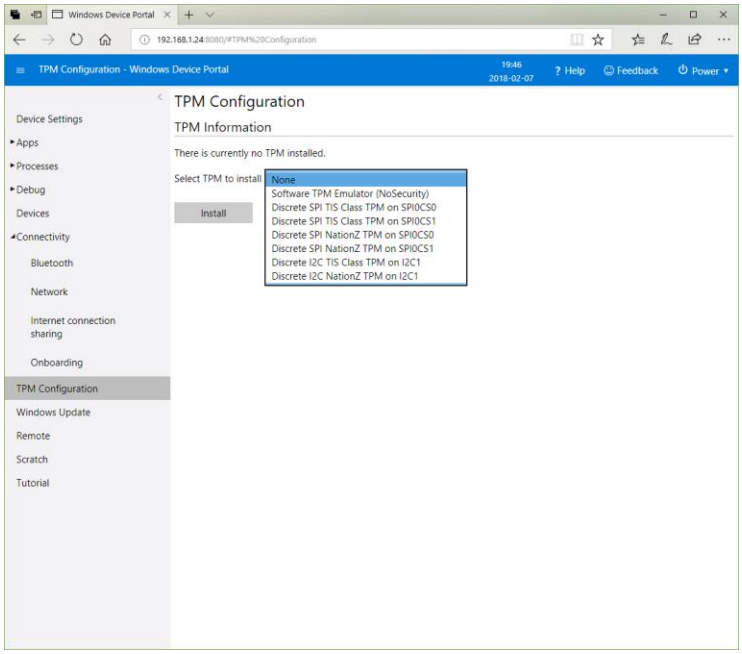

Figure 21: TPM Configuration view

RPI Windows 10 IoT sample project

Create simple Hello World application for Raspberry Pi board

To create simple Hello Word application under Windows 10 IOT Core the Visual Studio 2015 or higher is needed. Visual Studio must be installed with the Universal Windows Platform development extension.

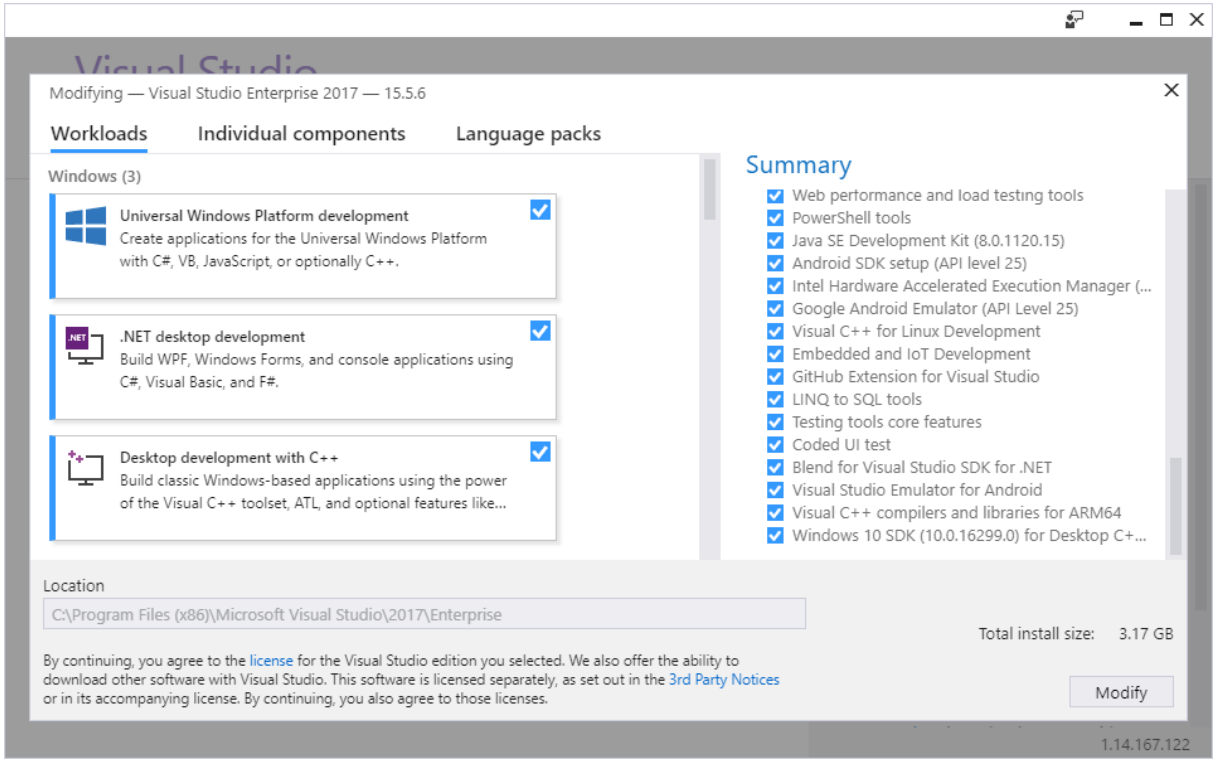

Figure 22: Visual Studio UWP development view

Step 1

Create new UWP project choosing the Windows Universal/Blank App Project.

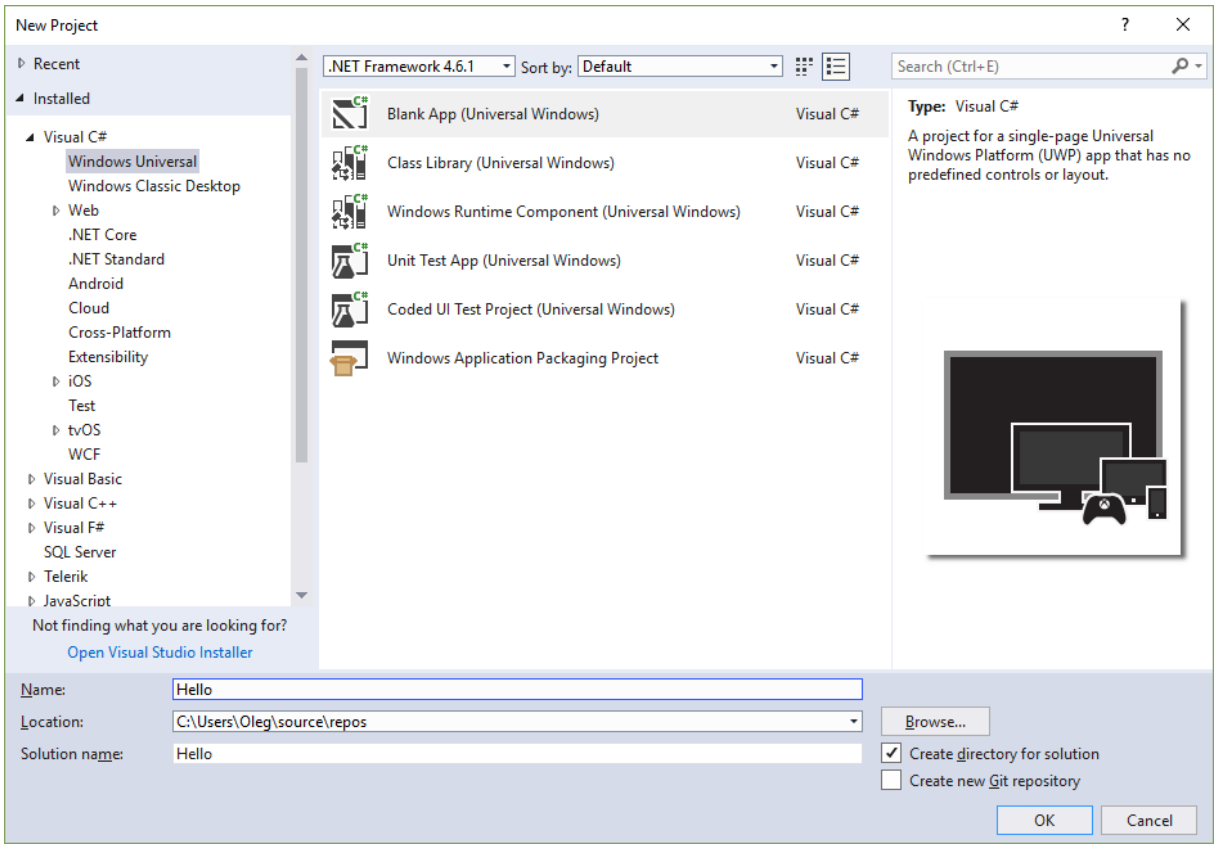

Figure 23: Create new UWP App

Step 2

Select target version (according to Raspberry Pi Windows 10 IOT Core build version)

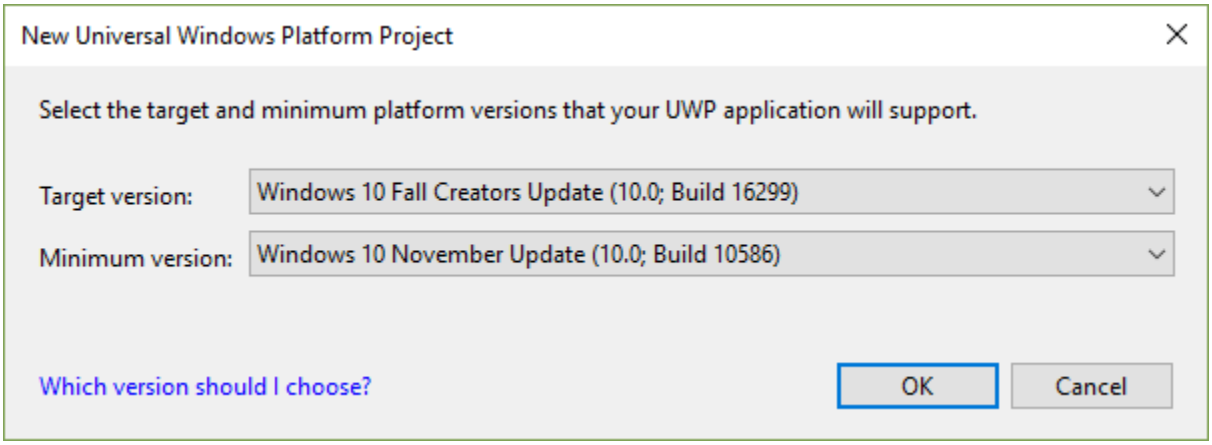

Figure 24: Select target version

Step 3

Create Hello solution and select target version

Step 4

Design the application screen modifying the MainPage.xaml file. To add different screen features use the Toolbox/All XAML controls.

Figure 25: Add TextBlock control

Step 5

Modify the MainPage.cs file content if you need controls events programming.

```
1   using System;
2   using System.Collections.Generic;
3   using System.IO;
4   using System.Linq;
5   using System.Runtime.InteropServices.WindowsRuntime;
6   using Windows.Foundation;
7   using Windows.Foundation.Collections;
8   using Windows.UI.Xaml;
9   using Windows.UI.Xaml.Controls;
10  using Windows.UI.Xaml.Controls.Primitives;
11  using Windows.UI.Xaml.Data;
12  using Windows.UI.Xaml.Input;
13  using Windows.UI.Xaml.Media;
14  using Windows.UI.Xaml.Navigation;
15
16  // The Blank Page item template is documented at https://go.microsoft.com/fwlink/?LinkId=402352&clcid=0x409
17
18  namespace Hello
19  {
20      /// <summary>
21      /// An empty page that can be used on its own or navigated to within a Frame.
22      /// </summary>
23      public sealed partial class MainPage : Page
24      {
25          public MainPage()
26          {
27              this.InitializeComponent();
28          }
29
30          private void TextBlock_SelectionChanged(object sender, RoutedEventArgs e)
31          {
32
33          }
34      }
35  }
36
```

Figure 26: Add TextBlock control code

Step 6

Compile and run Hello solution. Choosing the Solution Platform to the x86 user will be able to debug and run Hello application on the computers desktop emulator. This step is useful for program touchscreen design but is not capable of testing the sensors and controls programming due to software emulator restrictions.

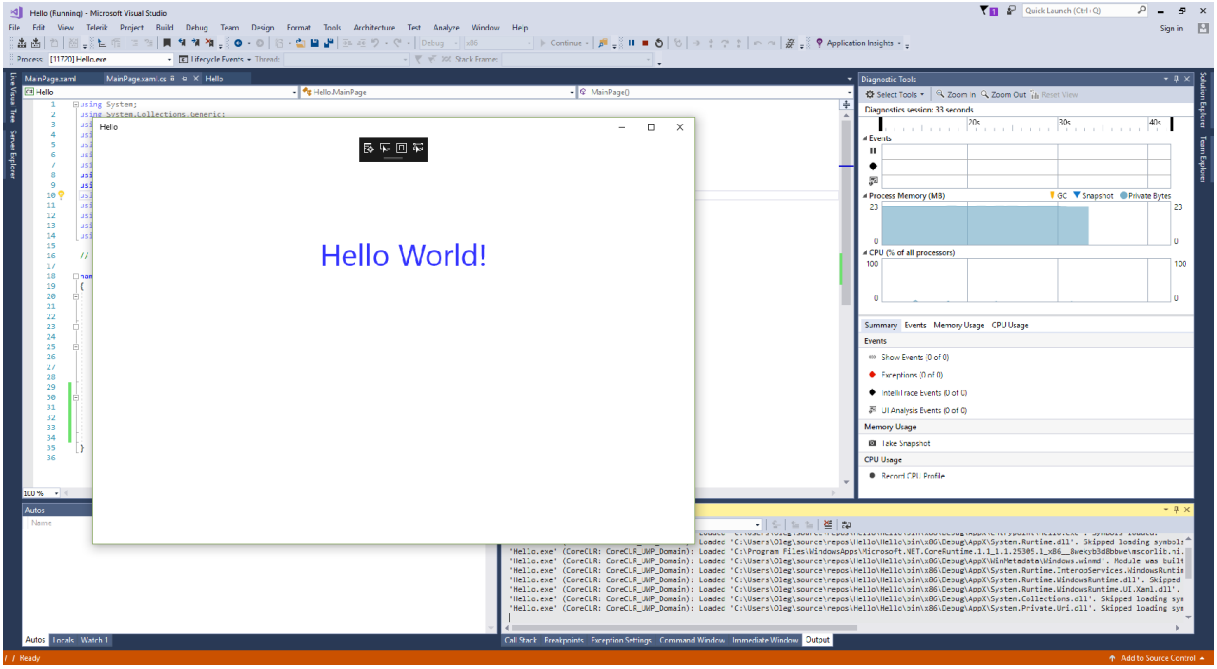

Figure 26: Run Hello solution

Software emulators aren't capable of simulating their behavior. To use sensors and controls instead, the Solution Platform must be changed to the ARM platform in the VC Solution Configuration property. To debug it application package must be then transferred to the real IoT device.

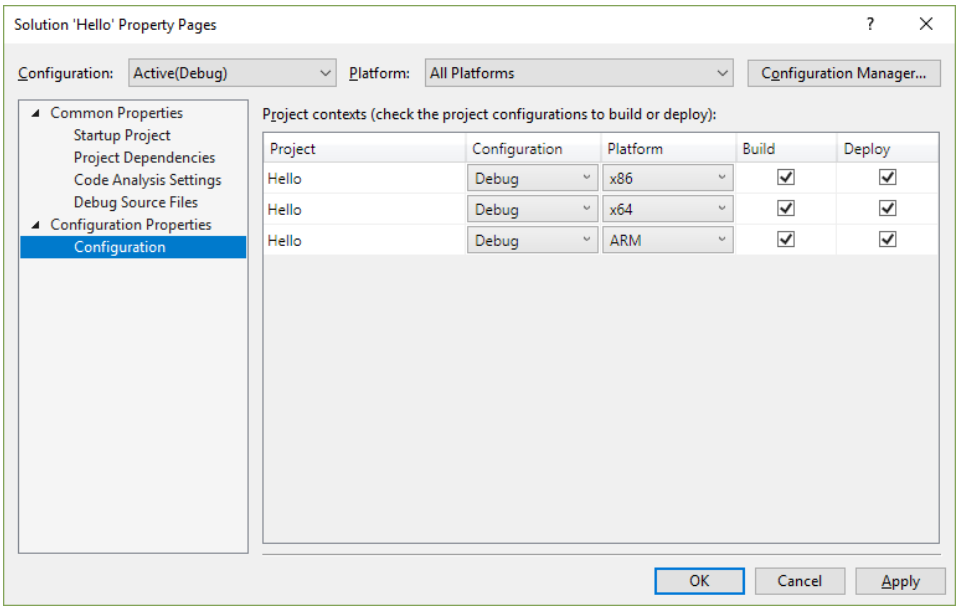

Figure 27: Select hardware platform for the solution

Step 7

To deploy and debug the application package on the real IoT device user must configure the debug application settings. In the Debug property page user must enter the proper Remote IoT device IP number.

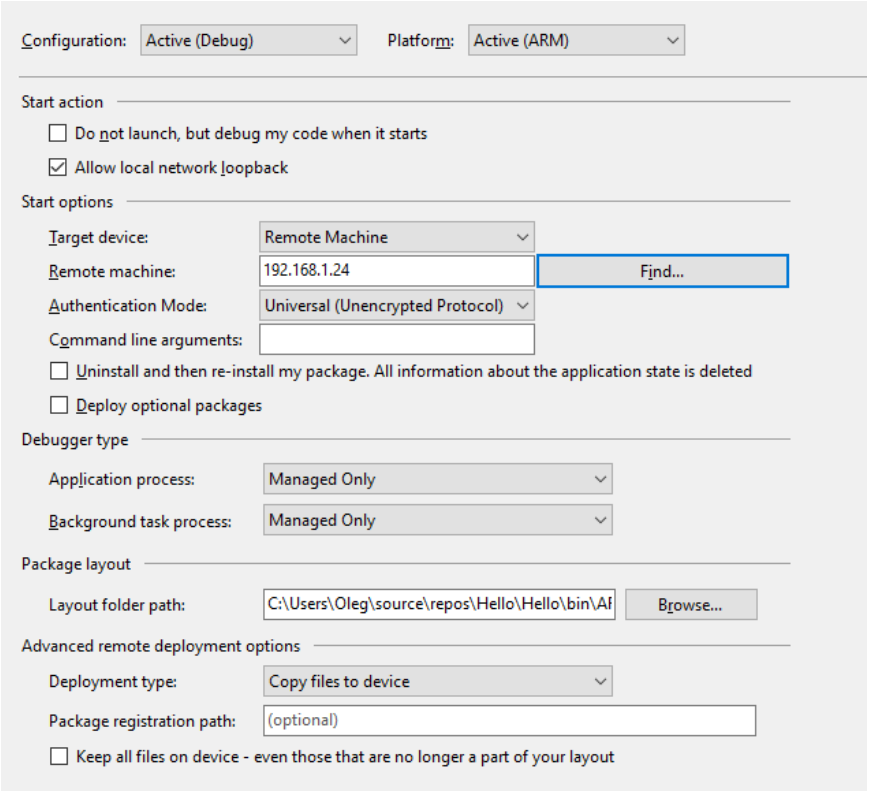

Figure 28: Set the Remote machine IP number

Step 8

Start debugging application and after deploy application package to the board SD card it will be displayed on the monitor.

Figure 29: Hello Application on the Raspberry Pi display

To Do

Follow the manual and create your own simple ***Hello World*** application for the Raspberry Pi developer board equipped with Windows 10 IoT Core OS system,

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